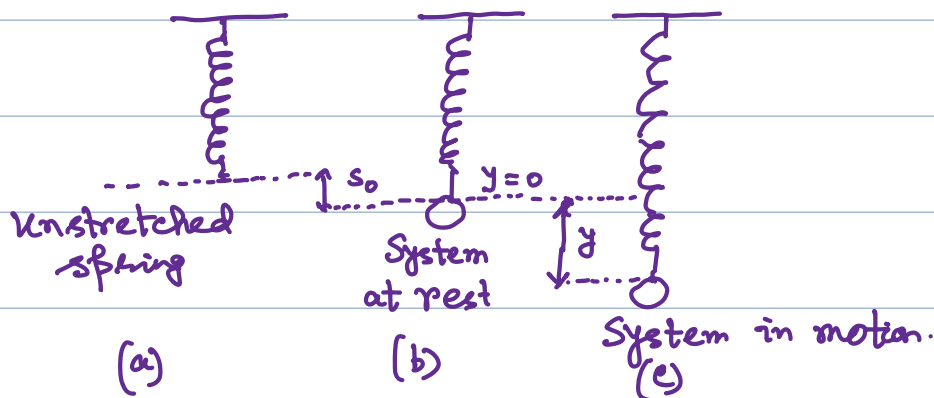


Free Oscillations (Undamped):

Free oscillations of a Mass-Spring System:



Mechanical mass-spring system

Let m be the mass of the ball.

Then Spring force $F = -ky$ $k > 0$ (Hooke's Law)

(always opposes the motion) (restoring force) ($k = \text{Spring Constant}$)

In (b), $ks_0 = mg$ ($- \text{Force present in the spring} = \text{Weight of the ball}$)

Now if a is the acceleration,

$$ma = -k(y + s_0) + mg = -ky$$

$$\Rightarrow y'' = -\frac{k}{m}y \Rightarrow y'' + \frac{k}{m}y = 0$$

Characteristic Equation: $\lambda^2 + \frac{k}{m} = 0 \Rightarrow \lambda = \pm i\sqrt{\frac{k}{m}}$

$$\Rightarrow \lambda = \pm i\omega_0$$

The general solution is $y = c_1 \cos(\omega_0 t) + c_2 \sin(\omega_0 t)$

This is called Harmonic Oscillation with

angular frequency $\omega_0 = \sqrt{\frac{k}{m}}$

Natural Frequency $f_0 = \frac{\omega_0}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$ (in Hz) (i.e. cycle/sec)

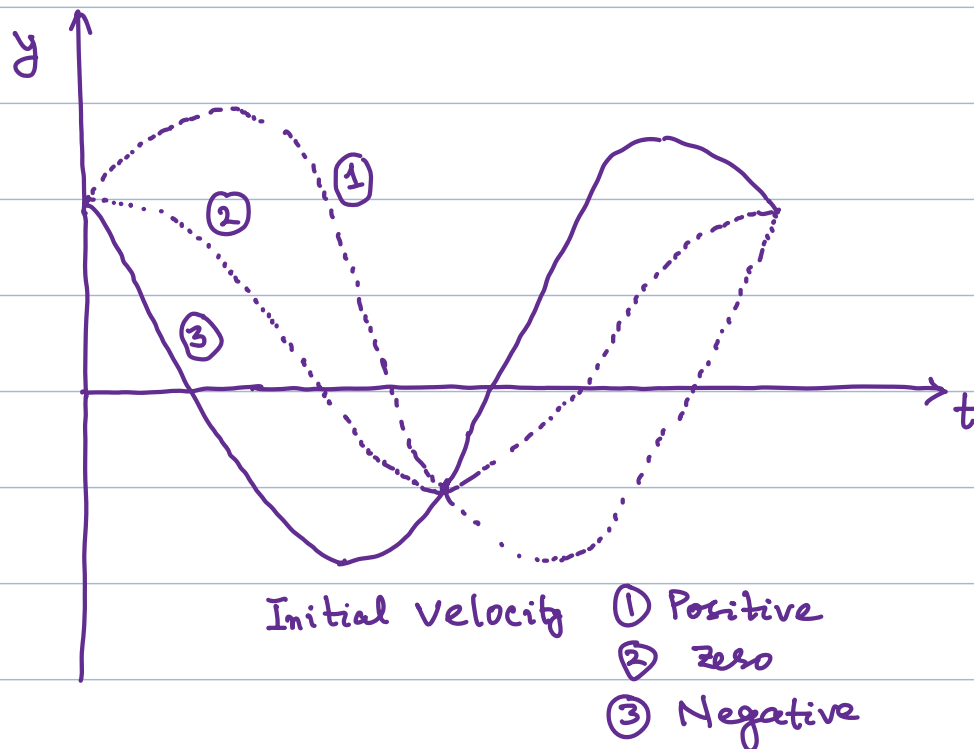
and Oscillation period $T_0 = \frac{1}{f_0} = 2\pi \sqrt{\frac{m}{k}}$ (in s)

Now we can replace $c_1 = C \cos \delta$ and $c_2 = C \sin \delta$,
for two constants C and δ .

$$\text{Then } y(t) = C (\cos \delta \cos(\omega_0 t) + \sin \delta \sin(\omega_0 t))$$

$$\Rightarrow y(t) = C \cos(\omega_0 t - \delta)$$

$$\text{Where } C = \sqrt{c_1^2 + c_2^2} \text{ and } \delta = \tan^{-1} \left(\frac{c_2}{c_1} \right)$$



Typical harmonic oscillations with same $y(0) = A$
and different initial velocities $y'(0) = \omega_0 B$ (Positive ①, Zero ②,
negative ③)

Ex: Weight = 98 N, $S_0 = 1.09$ m

How many cycles per minute will system execute?

$$f_0 = \frac{\omega_0}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

Now $ks_0 = mg \Rightarrow k \times 1.09 = 98 \Rightarrow k = 90$

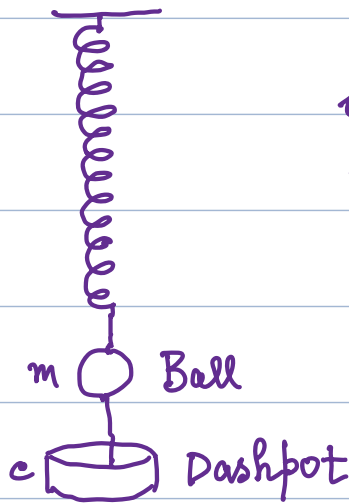
$m = 10$

Then $f_0 = \frac{1}{2\pi} \sqrt{\frac{90}{10}} = \frac{3}{2\pi} \approx 0.48 \text{ Hz } \left(\frac{1}{s}\right)$

In 1 minute, number of oscillations = $60 \times 0.48 \approx 29$ cycles/minute

Damped Oscillations:

Damped oscillations of a mass-spring system



There is a braking force present which can be modeled as $-c\dot{y}$ ($c > 0$)

Then $m\ddot{y} = -ky - c\dot{y}$ ($c = \text{damping constant}$)

$$\Rightarrow \ddot{y} = -\frac{k}{m}y - \frac{c}{m}\dot{y}$$

$$\Rightarrow \ddot{y} + \frac{c}{m}\dot{y} + \frac{k}{m}y = 0$$

The characteristic equation is

$$\lambda^2 + \frac{c}{m}\lambda + \frac{k}{m} = 0$$

$$\Rightarrow \lambda_1 = \frac{-\frac{c}{m} + \sqrt{\frac{c^2}{m^2} - \frac{4k}{m}}}{2} = \frac{-c + \sqrt{c^2 - 4mk}}{2m}$$

$$=: -\alpha + \beta = -(\alpha - \beta)$$

$$\text{and } \lambda_2 = \frac{-c - \sqrt{c^2 - 4mk}}{2m} =: -\alpha - \beta = -(\alpha + \beta)$$

where $\alpha = \frac{c}{2m}$ and $\beta = \frac{\sqrt{c^2 - 4mk}}{2m}$

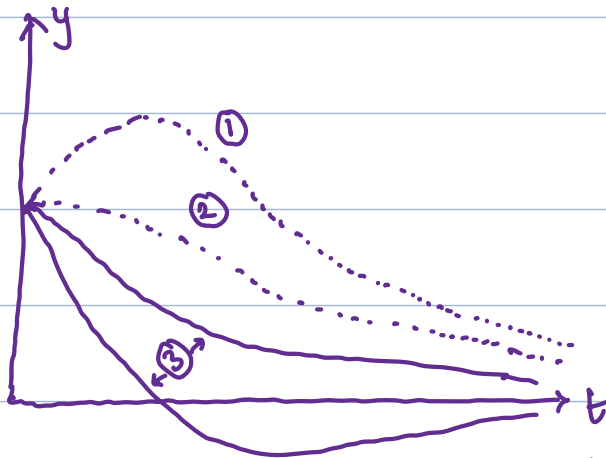
Now we need to consider three cases:

(1) $c^2 - 4mk > 0$ (overdamping):

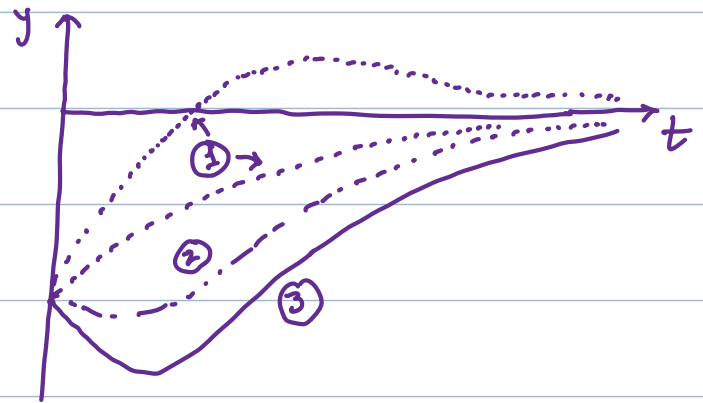
Then the roots of the characteristic equation are distinct and real.

$$y(t) = C_1 e^{-(\alpha-\beta)t} + C_2 e^{-(\alpha+\beta)t} \quad \left. \vphantom{y(t)} \right\}$$

- Note that $\alpha > \beta \Rightarrow y(t) \rightarrow 0$ as $t \rightarrow \infty$
- There is no oscillation, mass will pass the mean position ($y=0$) at most once.
- The damping takes out energy so quickly that the body does not oscillate.



(a) Positive initial displacement



(b) Negative initial displacement.

Initial velocity
① Positive
② Zero
③ Negative

(2) $c^2 - 4mk = 0$ (Critical damping):

$$\Rightarrow \alpha = \frac{c}{2m}, \quad \beta = 0$$

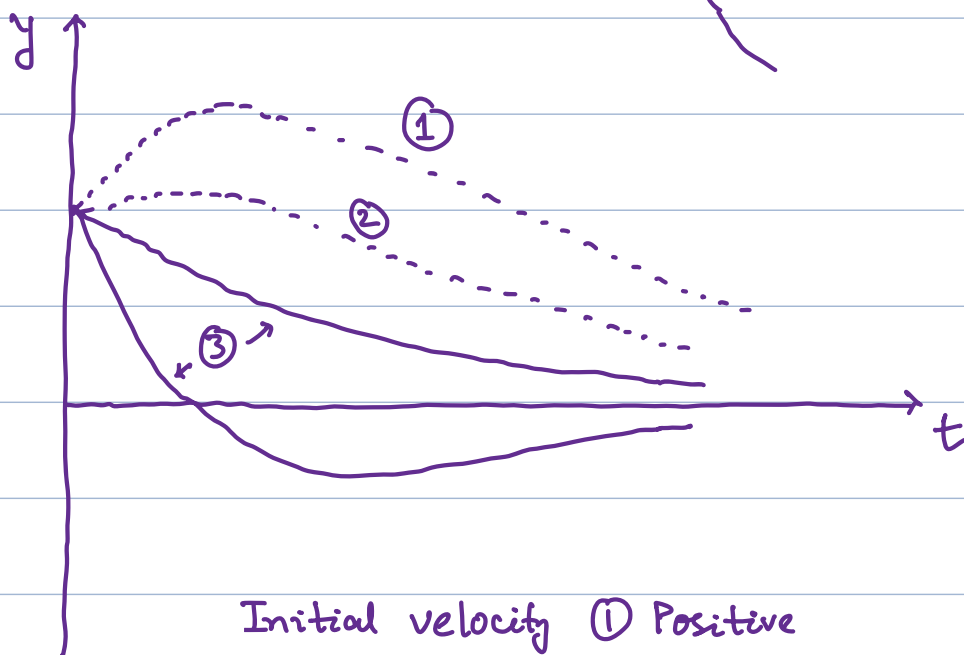
$$y(t) = (C_1 + C_2 t) e^{-\alpha t}$$

- $y(t) \rightarrow 0$ as $t \rightarrow \infty$

- No Oscillation

- Crosses mean position ($y=0$) at most once

$(e^{-\alpha t} > 0 \text{ and } c_1 + c_2 t = 0 \text{ for at most one } t > 0)$



Initial velocity ① Positive
② Zero
③ Negative

Note: Figures in overdamping and critical damping are very similar.

③ $c^2 - 4mk < 0$ (Under damping):

$$\beta = i \frac{1}{2m} \sqrt{4mk - c^2} = i \sqrt{\frac{k}{m} - \frac{c^2}{4m^2}} = i\omega^* \text{ (say)}$$

Note that $\omega^* \rightarrow \omega_0 = \sqrt{\frac{k}{m}}$ as $c \rightarrow 0$

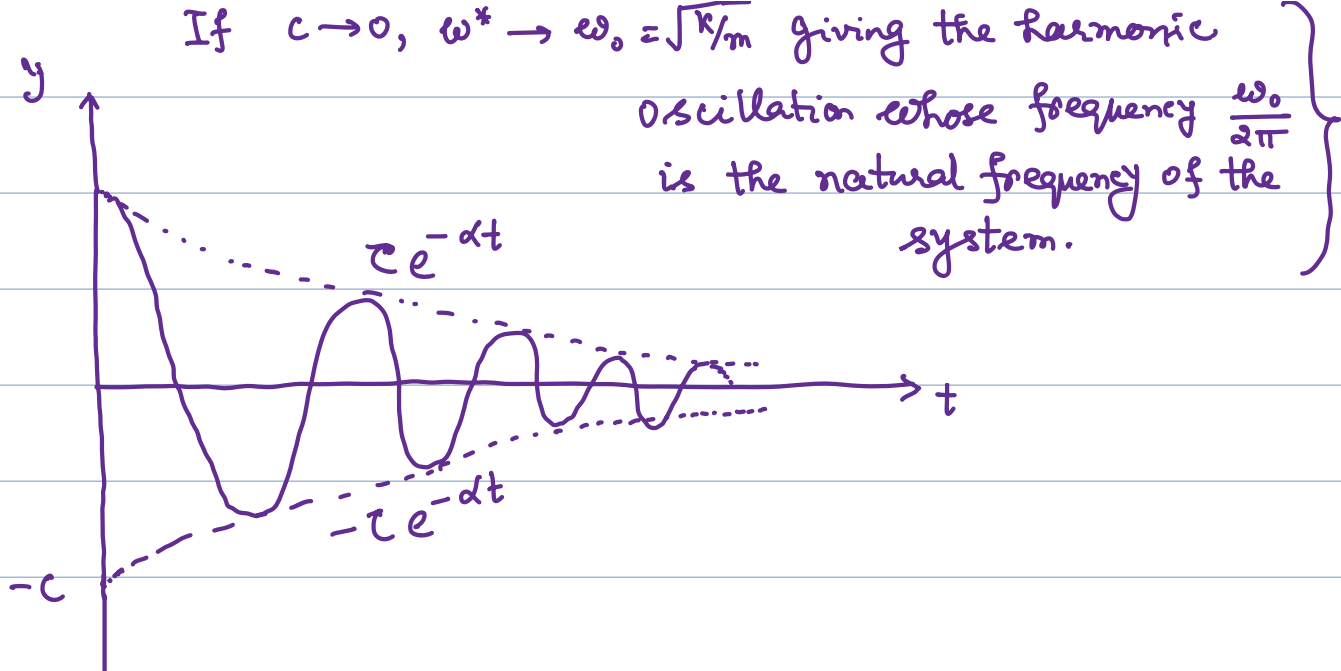
The general solution is $y(t) = e^{-\alpha t} (c_1 \cos(\omega^* t) + c_2 \sin(\omega^* t))$

$$\Rightarrow \boxed{y(t) = C e^{-\alpha t} \cos(\omega^* t - \delta)} \quad \text{where } C = \sqrt{c_1^2 + c_2^2} \text{ and } \delta = \tan^{-1}\left(\frac{c_2}{c_1}\right)$$

- Note that $y(t) \rightarrow 0$ as $t \rightarrow \infty$

- The motion is oscillatory.

- The frequency is $\omega^*/2\pi$ Hz. The smaller $c(>0)$ is, the larger is ω^* and the more rapid the oscillations become.



- The curve lies between the dashed curves $y = \tau e^{-\alpha t}$ and $y = -\tau e^{-\alpha t}$ touching them when $\omega^* t - \delta$ is an integer multiple of π

Euler-Cauchy Equations:

They are equations of the form

$$x^2 y'' + ax y' + by = 0$$

Let $y = x^m$ Then $y' = m x^{m-1}$, $y'' = m(m-1) x^{m-2}$

Then substituting we get

$$x^2 (m(m-1) x^{m-2}) + ax (m x^{m-1}) + b x^m = 0$$

$$\Rightarrow m(m-1) x^m + am x^m + b x^m = 0$$

$$\Rightarrow x^m (m(m-1) + am + b) = 0$$

$$\Rightarrow x^m (m^2 + (a-1)m + b) = 0$$

Hence x^m is a solution of ODE iff m is a solution of $m^2 + (a-1)m + b = 0$

The solutions are
$$\frac{-(a-1) \pm \sqrt{(a-1)^2 - 4b}}{2}$$

$$= \frac{1-a}{2} \pm \sqrt{\frac{(1-a)^2}{4} - b}$$

Case 1: Two distinct real roots m_1 and m_2 :

So, general solution is

$$y = c_1 x^{m_1} + c_2 x^{m_2}$$

where $m_1 = \frac{1-a}{2} + \sqrt{\frac{(1-a)^2}{4} - b}$

$$m_2 = \frac{1-a}{2} - \sqrt{\frac{(1-a)^2}{4} - b}$$

Ex:

$$4x^2 y'' + 4xy' - y = 0 \Rightarrow x^2 y'' + xy' - \frac{1}{4}y = 0$$

This is an Euler-Cauchy Equation with $a=1$, $b=-\frac{1}{4}$

So, $y = x^m$ is a solution iff $m^2 + (1-1)m - \frac{1}{4} = 0 \Rightarrow m^2 - \frac{1}{4} = 0$

$$\Rightarrow m^2 = \frac{1}{4}$$

$$\Rightarrow m = \frac{1}{2}, -\frac{1}{2}$$

So, the general solution is $y = c_1 x^{\frac{1}{2}} + c_2 x^{-\frac{1}{2}}$

$$\Rightarrow y = c_1 \sqrt{x} + \frac{c_2}{\sqrt{x}}$$

(It must be $x > 0$ for this solution to exist.)

Case 2: A real double root:

• A real double root occurs when $\frac{1}{4}(1-a)^2 - b = 0$

$$\Rightarrow b = \frac{1}{4}(1-a)^2$$

$$\text{Then } x^2 y'' + ax y' + by = 0 \Rightarrow x^2 y'' + ax y' + \frac{(1-a)^2}{4} y = 0$$

$$\Rightarrow y'' + \frac{a}{x} y' + \frac{(1-a)^2}{4x^2} y = 0$$

In this case one solution is $\left(m = \frac{1-a}{2} \right)$
 $y_1 = x^{\frac{1-a}{2}}$

By reduction of order

$$y_2 = y_1 \int \frac{1}{y_1^2} e^{-\int \frac{a}{x} dx} dx$$

$$= y_1 \int \frac{1}{y_1^2} e^{-a \ln|x|} dx$$

$$= y_1 \int \frac{1}{y_1^2} e^{\ln|x|^{-a}} dx = y_1 \int \frac{1}{y_1^2} |x|^{-a} dx$$

$$= x^{\frac{1-a}{2}} \int \frac{x^{-a}}{\left(x^{\frac{1-a}{2}}\right)^2} dx \quad (\text{for } x > 0)$$

$$= x^{\frac{1-a}{2}} \int \frac{x^{-a}}{x^{1-a}} dx = x^{\frac{1-a}{2}} \int \frac{dx}{x}$$

$$= x^{\frac{1-a}{2}} \ln x \quad (\text{for } x > 0)$$

Note that y_1 and y_2 are linearly independent.

So, the general solution is

$$y = C_1 y_1 + C_2 y_2 = \boxed{\left(C_1 + C_2 \ln x\right) x^{\frac{1-a}{2}}} \quad (\text{for } x > 0)$$

(3) The two roots m_1 and m_2 are complex

Let $m_1 = \alpha + i\omega$, $m_2 = \alpha - i\omega$ (say)

Two independent solutions are

$$y_1 = x^{\alpha + i\omega} = x^{\alpha} x^{i\omega} = x^{\alpha} (e^{\ln x})^{i\omega}$$

$$= x^{\alpha} e^{i\omega \ln x} = x^{\alpha} (\cos \omega(\ln x) + i \sin \omega(\ln x))$$

$$y_2 = x^{\alpha - i\omega} = x^{\alpha} x^{-i\omega} = x^{\alpha} (e^{-\ln x})^{i\omega}$$

$$= x^\alpha e^{-i\omega \ln x} = x^\alpha (\cos(\omega \ln x) - i \sin(\omega \ln x))$$

Two other independent solutions are

$$y_1^* = \frac{y_1 + y_2}{2} = x^\alpha \cos(\omega \ln x)$$

$$y_2^* = \frac{y_1 - y_2}{2i} = x^\alpha \sin(\omega \ln x)$$

So, the general solution is

$$y = C_1 y_1^* + C_2 y_2^* = x^\alpha \left(C_1 \cos(\omega \ln x) + C_2 \sin(\omega \ln x) \right)$$

Ex: $x^2 y'' + 3x y' + y = 0$ (Euler-Cauchy Equation with $a=3, b=1$)

So, the equation for m is $m^2 + (3-1)m + 1 = 0$

$$\Rightarrow m^2 + 2m + 1 = 0$$

$$\Rightarrow (m+1)^2 = 0$$

So, we have a real double root $m = -1$

Hence the general solution is

$$y(x) = (C_1 + C_2 \ln x) x^{-1} = \frac{(C_1 + C_2 \ln x)}{x}$$

(for $x > 0$)

Ex:

$$2x^2 y'' + 4xy' + 5y = 0$$

$$\Rightarrow x^2 y'' + 2xy' + \frac{5}{2}y = 0$$

(Euler-Cauchy equation with $a=2$, $b=\frac{5}{2}$)

So, the equation for m is $m^2 + (2-1)m + \frac{5}{2} = 0$

$$\Rightarrow m^2 + m + \frac{5}{2} = 0$$

$$\Rightarrow m^2 + m + 2.5 = 0$$

$$\text{So, } m = \frac{-1 \pm \sqrt{(1)^2 - 4(2.5)}}{2} = \frac{-1 \pm \sqrt{-9}}{2} = \frac{-1 \pm 3i}{2}$$

$$\Rightarrow m = -.5 \pm 1.5i$$

So, the roots are complex and the general

solution is $y(x) = x^{-.5} [c_1 \cos(1.5 \ln x) + c_2 \sin(1.5 \ln x)]$

Ex: Find the electrostatic potential

$v = v(r)$ between two concentric

sphere of radius $r_1 = 5\text{cm}$ and $r_2 = 10\text{cm}$

kept at potential $v_1 = 110\text{V}$ and $v_2 = 0$

Physical information: $v(r)$ is a solution

of $rv'' + 2v' = 0$ where $v' = \frac{dv}{dr}$

$$r v'' + 2v' = 0$$

Multiplying by r , we get

$$r^2 v'' + 2r v' = 0$$

$v = r^m$ will be a solution iff $m^2 + (2-1)m = 0$

$$\Rightarrow m^2 + m = 0 \Rightarrow m = 0, -1$$

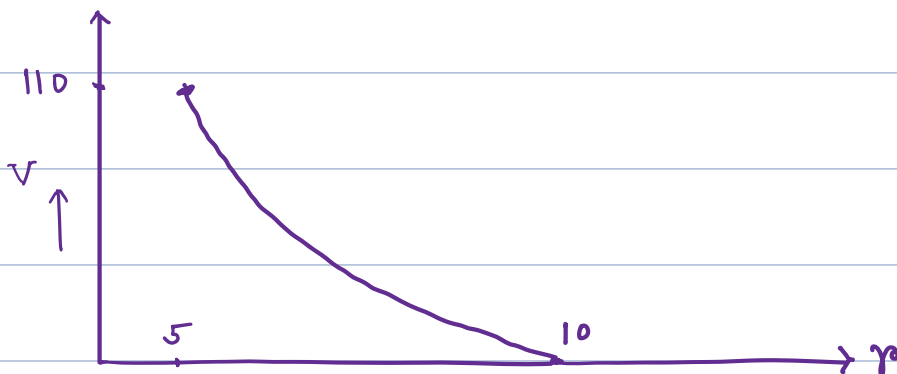
So, the general solution is

$$v = c_1 r^0 + c_2 r^{-1} = c_1 + \frac{c_2}{r}$$

$$\left. \begin{aligned} v(5) = 110 &\Rightarrow c_1 + \frac{c_2}{5} = 110 \\ v(10) = 0 &\Rightarrow c_1 + \frac{c_2}{10} = 0 \end{aligned} \right\}$$

$$c_2 = 1100, c_1 = -110$$

So, $v = -110 + \frac{1100}{r}$



Ex: Solve: $x^2 y'' - 2x y' + 2.25 y = 0$, Given: $y(1) = 2.2$, $y'(1) = 2.5$
(Euler-Cauchy Equation with $a = -2$, $b = 2.25$)

$y = x^m$ is a solution iff $m^2 + (-2-1)m + 2.25 = 0$

$$\Rightarrow m^2 - 3m + 2.25 = 0$$

$$\Rightarrow (m - 1.5)^2 = 0 \Rightarrow m = 1.5$$

So, it has a real double root at $m = 1.5$

The general solution is $y(x) = (c_1 + c_2 \ln x) x^{1.5}$ ($x > 0$)

$$\text{Now } y(1) = 2.2 \Rightarrow [c_1 + c_2(0)](1)^{1.5} = 2.2$$

$$\Rightarrow c_1 = 2.2$$

$$\begin{aligned} \text{Now } y'(x) &= (1.5)x^{.5} (c_1 + c_2 \ln x) + c_2 \frac{1}{x} \cdot x^{1.5} \\ &= 1.5 (c_1 + c_2 \ln x) x^{.5} + c_2 x^{.5} \end{aligned}$$

$$\text{Now } y'(1) = 2.5 \Rightarrow 2.5 = (1.5)(2.2) + c_2$$

$$\Rightarrow 2.5 = 3.3 + c_2$$

$$\Rightarrow c_2 = 2.5 - 3.3 = -.8$$

So, the particular solution is $y(x) = (2.2 - .8 \ln x) x^{1.5}$